

PROBLEMA № 04

DISEÑAR UNA ZAPATA AISLADA CON LAS SIGUIENTES CONSIDERACIONES TÉCNICAS Y ESTRUCTURALES.

$$P_{CH} = 125 \text{ ton}$$

$$M_{CHX} = 2.75 \text{ ton-m}$$

$$M_{CH_4} = 2.45 \text{ ton-m}$$

$$P_{CV} = 65 \text{ ton}$$

$$M_{cyx} = 1.65 \text{ ton-m}$$

$$M_{crg} = 1.25 \text{ ton-m}$$

$P_{CS} = 0 \text{ ton}$

$$M_{CSx} = 4.15 \text{ ton-m}$$

$$M_{csy} = 6.55 \text{ ton-m.}$$

$$\Delta T = 2.50 \text{ } ^\circ\text{C}$$

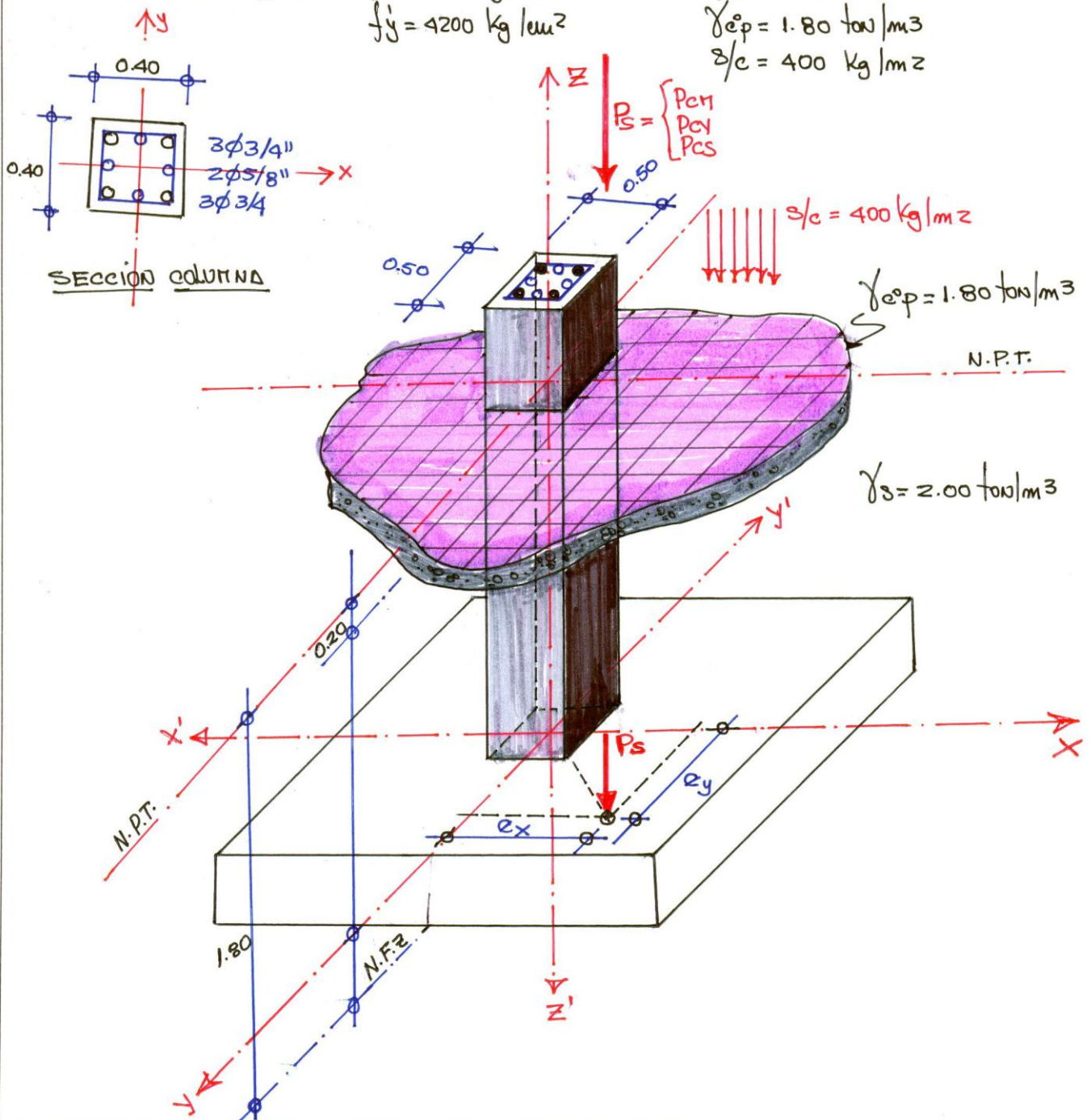
$$f_c' = 210 \text{ kg/cm}^2$$

$$\gamma_s = 2.00 \text{ ton/m}^3$$

$$\dot{\gamma} = 4200 \text{ kg/cm}^2$$

$$\gamma_{ep} = 1.80 \text{ ton/m}^3$$

$$8/c = 400 \text{ Kg/m}^2$$



A: DIMENSIONAMIENTO DE LA ZAPATAPERALTE DE LA ZAPATA

CONSIDERACIONES NORMATIVAS

$$A_{seol}: 6\phi 3/4" + 4\phi 5/8 \Rightarrow \phi 3/4 \left\{ \begin{array}{l} d_b = 1.905 \text{ cm.} \\ A_b = 2.84 \text{ cm}^2 \end{array} \right.$$

$$\left. \begin{array}{l} l_d = 0.08 d_b f_y / \sqrt{f'_c} = 44.170 \\ l_d = 0.004 d_b f_y = 32.004 \\ l_d \geq 20 \text{ cm.} \end{array} \right\} \begin{array}{l} l_d = 44.17 \text{ cm.} \\ l_d = 45 \text{ cm.} \end{array}$$

$$\circ d = l_d + 10 \text{ cm} \Rightarrow d = 55.00 \text{ cm.} \quad h = d + 10 \text{ cm} \Rightarrow h = 65.00 \text{ cm.}$$

CAPACIDAD PORTANTE NETA DEL TERRENO (q_e)

$$q_e = q_t - (\gamma_{ce} \times h_z) - (\gamma_s \times h_s) - (\gamma_{cp} \times h_p) - q/c$$

$$= 25.00 - (2.40 \times 0.65) - (2.00 \times 0.95) - (1.80 \times 0.20) - 0.40$$

$$q_e = 20.78 \text{ ton/m}^2 = 2.078 \text{ kg/cm}^2$$

AREA DE LA ZAPATA

$$\boxed{A = \frac{P_t}{q_e}} \Rightarrow A = \frac{P_{cm} + P_{cy} + P_{cs}}{q_e} \Rightarrow A = \frac{125 + 65 + 0}{20.78} \Rightarrow A = 9.143 \text{ m}^2$$

$$\circ \text{Columna } f/\text{columna} \Rightarrow \text{Zapata cuadrada } B = \sqrt{A} \Rightarrow B = 3.024 \text{ m}$$

$$\circ B_x = 3.00, B_y = 3.00$$

$$A = 9.00 \text{ m}^2$$

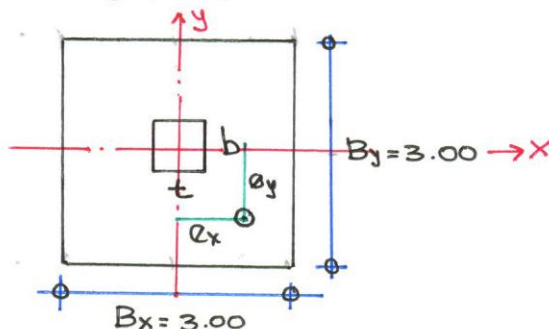
VERIFICACION DE PRESIONES: $\sigma_{max} < \sigma_t$ A-1: ANALISIS ESTATICOCARGAS DE SERVICIO:

$$P_s = P_{cm} + P_{cy} \Rightarrow P_s = 125 + 65 \Rightarrow P_s = 190.00 \text{ ton.}$$

MOMENTOS DE SERVICIO:

$$M_x = M_{cmx} + M_{cyy} \Rightarrow M_x = 2.75 + 1.65 \Rightarrow M_x = 4.40 \text{ ton-m.}$$

$$M_y = M_{cmy} + M_{cyy} \Rightarrow M_y = 2.45 + 1.25 \Rightarrow M_y = 3.70 \text{ ton-m.}$$



DIRECCION : X-X

$$\frac{B_x}{6} = \frac{3.00}{6} \Rightarrow \frac{B_x}{6} = 0.50 \text{ m.}$$

$$Q_x = \frac{M_x}{P_s} \Rightarrow Q_x = \frac{4.40}{190} \Rightarrow Q_x = 0.023 \text{ m.}$$

$\Rightarrow \frac{B_x}{6} < Q_x \Rightarrow$ Presión de contacto porra trapezoidal

$$\sigma_{\max} = \frac{P_s}{A} + \frac{M_x c}{I}$$

$$P_s = 190.00 \text{ ton}$$

$$A = 9.00 \text{ m}^2$$

$$M_x = 4.40 \text{ ton-m}$$

$$B = 3.00 \text{ m.}$$

$$C = \frac{B_x}{2} \Rightarrow C = 1.50 \text{ m}$$

$$I = \frac{b_y b_x^3}{12} \Rightarrow I = 6.75 \text{ m}^4$$

$$\sigma_{\max} = \frac{190}{9.00} + \frac{4.40(1.50)}{6.75} \Rightarrow \sigma_{\max} = 22.089 \text{ ton/m}^2 < \sigma_t = 25 \text{ ton/m}^2$$

(CONFORME.)

DIRECCION : Y-Y

$$M_y = 3.70 \text{ ton-m}$$

$$\frac{B_y}{6} = \frac{3.00}{6} \Rightarrow \frac{B_y}{6} = 0.50 \text{ m.}$$

$$Q_y = \frac{M_y}{P_s} \Rightarrow Q_y = \frac{3.70}{190} \Rightarrow Q_y = 0.019 \text{ m}$$

$\Rightarrow \frac{B_y}{6} < Q_y \Rightarrow$ Presión de contacto porra trapezoidal

$$\sigma_{\max} = \frac{P_s}{A} + \frac{M_y c}{I}$$

$$\sigma_{\max} = \frac{190}{9.00} + \frac{3.70(1.50)}{6.75} \Rightarrow \sigma_{\max} = 21.933 \text{ ton/m}^2 < \sigma_t = 25 \text{ ton/m}^2$$

(CONFORME.)

A.2 : ANALISIS SISTICO► CARGAS DE SERVICIO

$$P_s = P_{\text{per}} + P_{\text{ex}} + P_{\text{cs}} \Rightarrow P_s = 125 + 65 + 0 \Rightarrow P_s = 190 \text{ ton.}$$

► MOMENTOS DE SERVICIO

$$M_x = M_{\text{extx}} + M_{\text{evx}} + M_{\text{esx}} \Rightarrow M_x = 2.75 + 1.65 + 4.15 \Rightarrow M_x = 8.55 \text{ ton-m}$$

$$M_y = M_{\text{exty}} + M_{\text{evy}} + M_{\text{esy}} \Rightarrow M_y = 2.45 + 1.25 + 6.55 \Rightarrow M_y = 10.25 \text{ ton-m.}$$

DIRECCION : X-X

$$Q_x = \frac{M_x}{P_s} \Rightarrow Q_x = \frac{8.55}{190} \Rightarrow Q_x = 0.045 \quad \frac{B_x}{6} = 0.50$$

Presión de contacto porra trapezoidal

$$\sigma_{\max} = \frac{P_s}{A} + \frac{M_x c}{I}$$

$$\sigma_{\max} = \frac{190}{9.00} + \frac{8.55(1.50)}{6.75} \Rightarrow \sigma_{\max} = 23.011 \text{ ton/m}^2 < \sigma_t$$

(CONFORME.)

DIRECCION : y-y

$$Q_y = \frac{M_y}{P_s} \Rightarrow Q_y = \frac{10.25}{190.00} \Rightarrow Q_y = 0.054 < \frac{1}{6} \quad \text{Presión de contacto forma trapezoidal}$$

$$\sigma_{\max} = \frac{P_s}{A} + \frac{M_y(c)}{I}$$

$$\sigma_{\max} = \frac{190}{9.00} + \frac{10.25(1.50)}{6.75} \Rightarrow \sigma_{\max} = 23.389 \text{ ton/m}^2 < \sigma_g = 25 \text{ ton/m}^2 \quad (\text{CONFORME})$$

CARGAS DE DISEÑO : (P_u, M_u)REACCION AMPLIFICADA DEL SUELO :ANALISIS ESTADICO :

$$P_u = 1.40 P_{em} + 1.70 P_{ev} \Rightarrow P_u = 1.40(125) + 1.70(65) \Rightarrow P_u = 285.50 \text{ ton.}$$

$$M_{ux} = 1.40 M_{emx} + 1.70 M_{evx} \Rightarrow M_{ux} = 1.40(2.75) + 1.70(1.65) \Rightarrow M_{ux} = 6.655 \text{ ton-m}$$

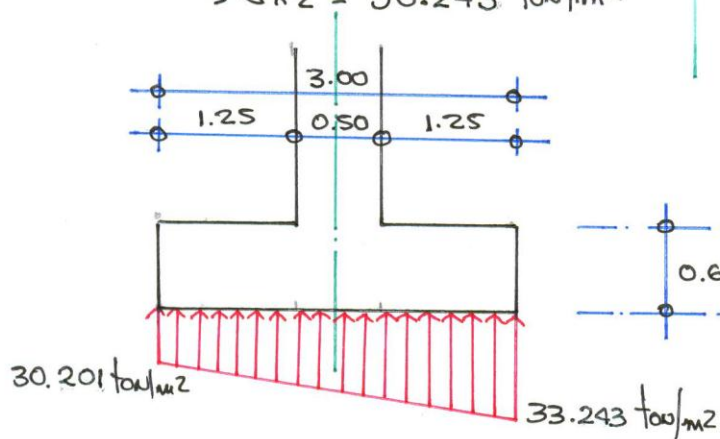
$$M_{uy} = 1.40 M_{emy} + 1.70 M_{evy} \Rightarrow M_{uy} = 1.40(2.45) + 1.70(1.25) \Rightarrow M_{uy} = 5.555 \text{ ton-m}$$

DIRECCION : x-x $\frac{B_x}{6} = \frac{B_y}{6} = 0.50$

$$Q_x = \frac{M_{ux}}{P_u} \Rightarrow Q_x = \frac{6.655}{285.50} \Rightarrow Q_x = 0.023 < \frac{B_x}{6} \quad (\text{CONFORME})$$

$$\sigma_{x(1,2)} = \frac{P_u}{A} \pm \frac{M_{ux}(c)}{I} = \frac{285}{9.00} \pm \frac{6.655(1.50)}{6.75}$$

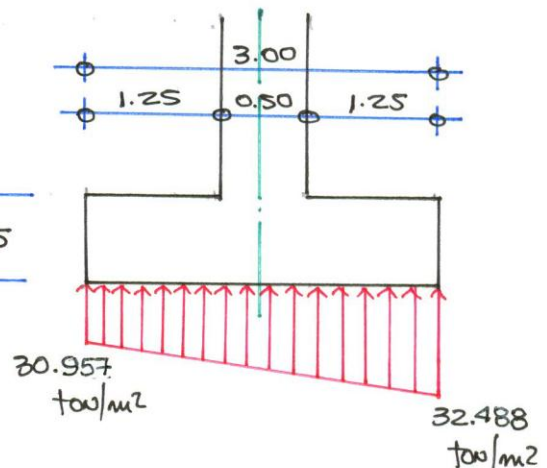
$$\sigma_{x(1,2)} \begin{cases} \sigma_{x1} = 33.201 \text{ ton/m}^2 \\ \sigma_{x2} = 30.243 \text{ ton/m}^2 \end{cases}$$

DIRECCION : y-y

$$Q_y = \frac{M_{uy}}{P_u} = \frac{5.555}{285.5} \Rightarrow Q_y = 0.019 < \frac{B_y}{6} \quad (\text{CONFORME})$$

$$\sigma_{y(1,2)} = \frac{P_u}{A} \pm \frac{M_{uy}(c)}{I} = \frac{285}{9.00} \pm \frac{5.555(1.50)}{6.75}$$

$$\sigma_{y(1,2)} \begin{cases} \sigma_{y1} = 32.957 \text{ ton/m}^2 \\ \sigma_{y2} = 30.488 \text{ ton/m}^2 \end{cases}$$



ANÁLISIS SISTICO:

$$P_{CH} = 125 \text{ ton}$$

$$P_{CV} = 65 \text{ ton}$$

$$P_{CS} = \emptyset$$

$$M_{CHX} = 2.75 \text{ ton-m}$$

$$M_{CVX} = 1.65 \text{ ton-m}$$

$$M_{CSX} = 4.65 \text{ ton-m}$$

$$M_{CHY} = 2.45 \text{ ton-m}$$

$$M_{CVY} = 1.25 \text{ ton-m}$$

$$M_{CSY} = 6.55 \text{ ton-m}$$

COMBINACIONES DE CARGA:

$$U = 1.25 (CH + CV) \pm CS$$

$$U = 0.90 CH \pm CS$$

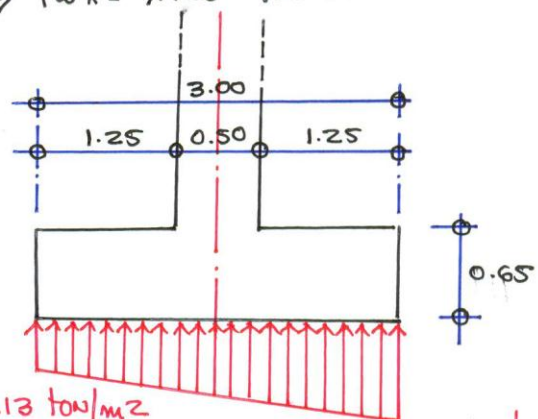
DIRECCION: X-X

$$P_U: \begin{cases} P_U = 1.25 (125 + 65) + 0 \Rightarrow P_U = 237.50 \text{ ton} \checkmark \\ P_U = 0.90 (125) + 0 \Rightarrow P_U = 112.50 \text{ ton} \end{cases}$$

$$M_{UX}: \begin{cases} M_{UX} = 1.25 (2.75 + 1.65) + 4.65 \Rightarrow M_{UX} = 10.15 \text{ ton-m} \checkmark \\ M_{UX} = 0.90 (2.75) + 4.65 \Rightarrow M_{UX} = 7.125 \text{ ton-m} \end{cases}$$

$$\sigma_{x1,2} = \frac{P_U}{A} \pm \frac{M_{UX} c}{I}$$

$$\sigma_{x1,2} \begin{cases} \sigma_{x1} = 28.64 \text{ ton/m}^2 \\ \sigma_{x2} = 24.13 \text{ ton/m}^2 \end{cases}$$

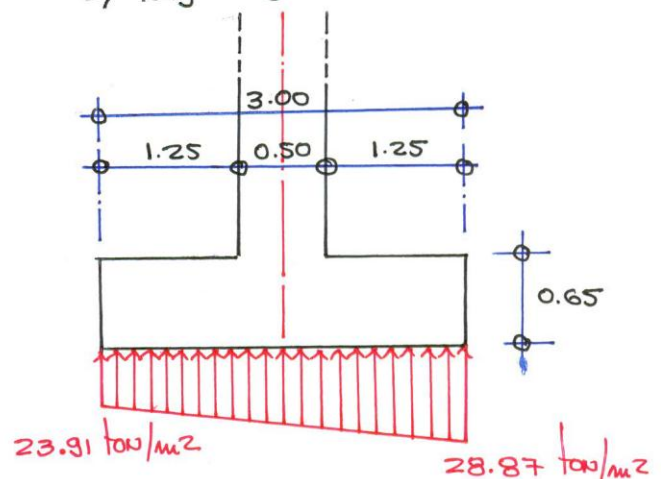
DIRECCION: Y-Y

$$P_U: \begin{cases} P_U = 1.25 (125 + 65) + 0 \Rightarrow P_U = 237.50 \text{ ton} \checkmark \\ P_U = 0.90 (125) + 0 \Rightarrow P_U = 112.50 \text{ ton} \end{cases}$$

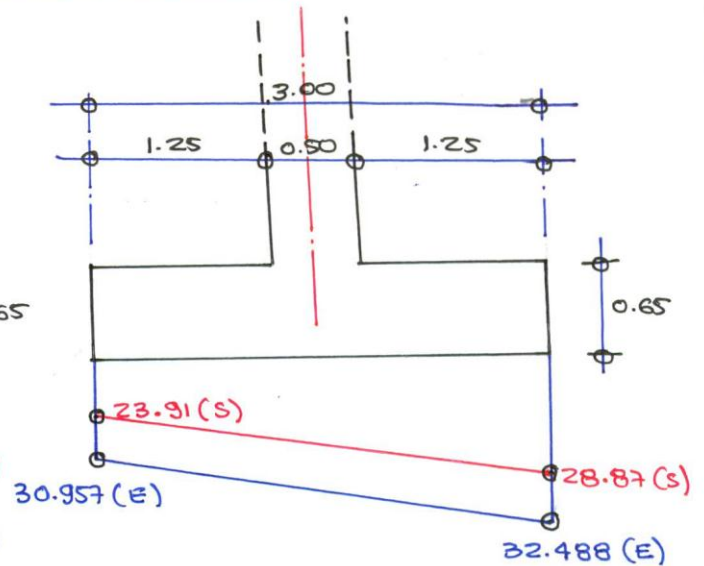
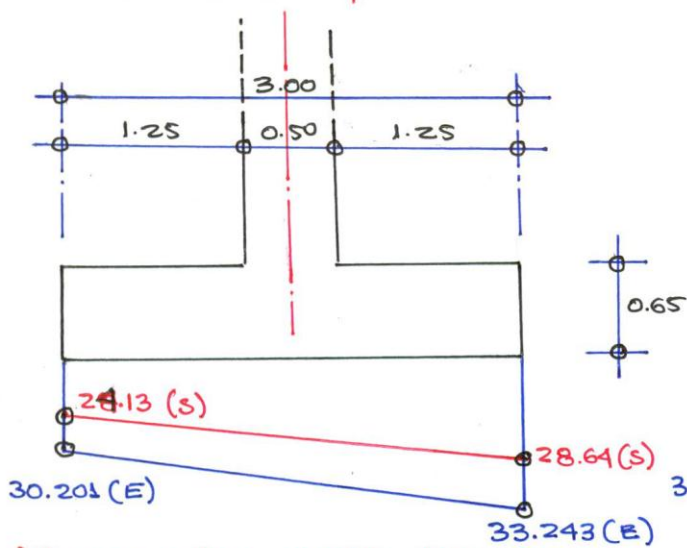
$$M_{UY}: \begin{cases} M_{UY} = 1.25 (2.45 + 1.25) + 6.55 \Rightarrow M_{UY} = 11.175 \text{ ton-m} \checkmark \\ M_{UY} = 0.90 (2.45) + 6.55 \Rightarrow M_{UY} = 8.755 \text{ ton-m} \end{cases}$$

$$\sigma_{y1,2} = \frac{P_U}{A} \pm \frac{M_{UY} c}{I}$$

$$\sigma_{y1,2} \begin{cases} \sigma_{y1} = 28.87 \text{ ton/m}^2 \\ \sigma_{y2} = 23.91 \text{ ton/m}^2 \end{cases}$$



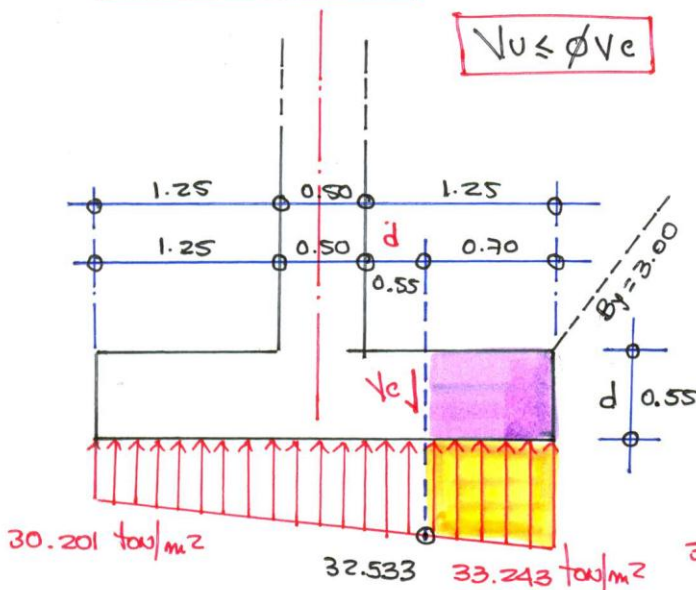
RESUMEN DE PRESIONES (ESTATICO - SISTICO)



VERIFICACION CORTE-FLEXION

DIRECCION X-X

$$V_u \leq \phi V_c$$



$$V_u = \left[\frac{(33.243 + 32.533) \cdot 0.70}{2} \right] 3.00$$

$$V_u = 69.065 \text{ ton} = 69064.800 \text{ kg}$$

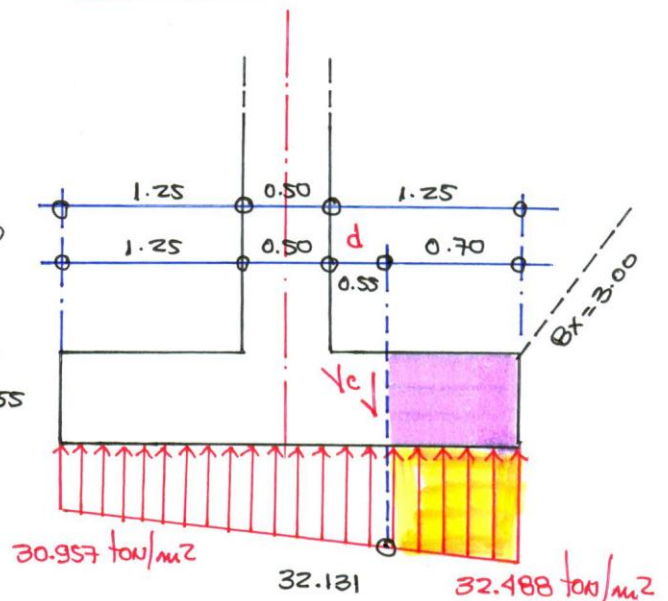
$$\phi V_c = \phi 0.53 \sqrt{f_c} B_y d$$

$$\phi V_c = 0.85 \cdot 0.53 \cdot \sqrt{210} \cdot 300 \cdot 55$$

$$\phi V_c = 107.718 \text{ ton} = 107718.00 \text{ kg}$$

CONFORME: $V_u < \phi V_c$

DIRECCION Y-Y



$$V_u = \left[\frac{(32.488 + 32.131) \cdot 0.70}{2} \right] 3.00$$

$$V_u = 67.850 \text{ ton} = 67849.95 \text{ kg}$$

$$\phi V_c = \phi 0.53 \sqrt{f_c} B_x d$$

$$\phi V_c = 0.85 \cdot 0.53 \cdot \sqrt{210} \cdot 200 \cdot 55$$

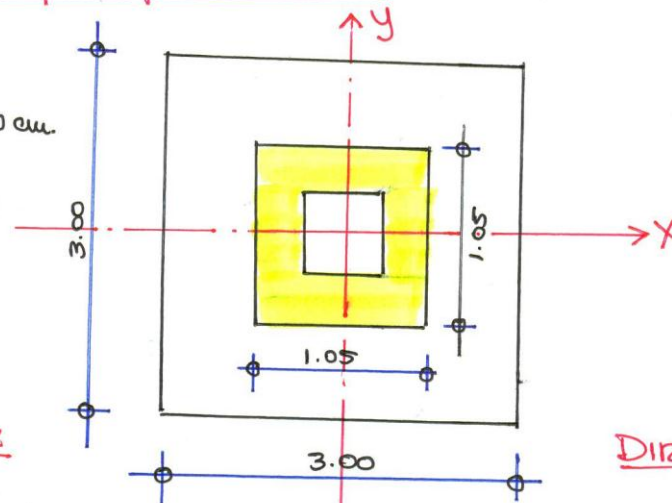
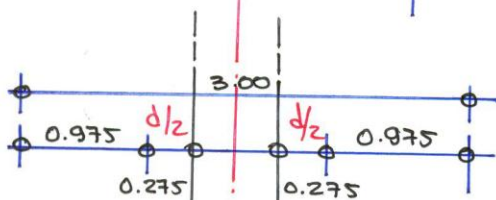
$$\phi V_c = 107.718 \text{ ton} = 107718.00 \text{ kg}$$

CONFORME: $V_u < \phi V_c$

VERIFICACIÓN POR FUNZONAMIENTO:

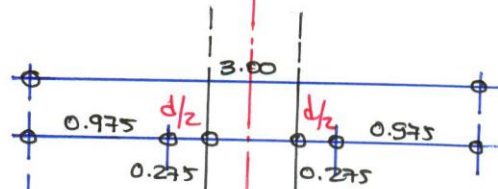
$$b_0 = 4(1.05) \Rightarrow b = 420 \text{ cm}$$

$$\beta_0 = \frac{0.50}{0.50} \Rightarrow \beta_0 = 1$$

DIRECCION: X-X

$$V_u \leq \phi V_c$$

$$d = 0.55$$

DIRECCION: Y-Y

$$V_u = \left[\frac{(33.243 + 30.201) \cdot 3.00}{2} \right] 3.00 - \left[\frac{(32.254 + 31.190) \cdot 1.05}{2} \right] 1.05$$

$$V_u = 250.524 \text{ ton} = 250524.495 \text{ kg}$$

$$\phi V_c = \phi \left(0.53 + \frac{1.10}{\beta_0} \right) \sqrt{f_c} b_0 d$$

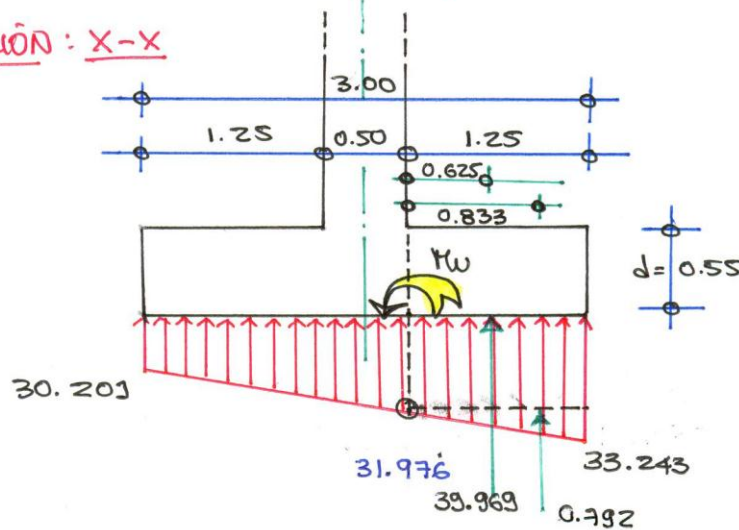
$$\phi V_c = 0.85 \left(0.53 + \frac{1.10}{1} \right) \sqrt{210} 420 \cdot 55 \Rightarrow \phi V_c = 463.797 \text{ ton} = 463797.237 \text{ kg}$$

DIRECCION: Y-Y

$$V_u = \left[\frac{(32.488 + 30.957) \cdot 3.00}{2} \right] 3.00 - \left[\frac{(31.99 + 31.455) \cdot 1.05}{2} \right] 1.05$$

$$V_u = 250.528 \text{ ton} = 250528.444 \text{ kg}$$

$$\therefore V_{ux}, V_{uy} < \phi V_c \Rightarrow \text{CONFORME}$$

DISEÑO DE REFUERZO POR FLEXIÓNDIRECCIÓN: X-X

$$f_c = 210 \text{ kg/cm}^2$$

$$f_y = 4200 \text{ kg/cm}^2$$

$$M_w = 39.969 (0.625) + 0.792 (0.833) \Rightarrow M_w = 25.640 \text{ ton-m}$$

DATOS DE DISEÑO: $M_w = 26.135 \text{ ton-m}$, $b = 100 \text{ cm}$, $d = 55 \text{ cm}$, $f_c = 210 \text{ kg/cm}^2$, $f_y = 4200 \text{ kg/cm}^2$

$$\alpha = \frac{A_s f_y}{0.85 f_c b}$$

$$M_w = \phi A_s f_y (d - a/2)$$

$$\rho_{min} = 0.70 \sqrt{f_c} / f_y \Rightarrow \rho_{min} = 0.00242 \Rightarrow A_s = \rho_{min} b d \Rightarrow A_{smin} = 13.284 \text{ cm}^2$$

$$\rho_b = 0.85 \beta \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow \rho_b = 0.02125 \Rightarrow A_{sb} = \rho_b b d \Rightarrow A_{sb} = 116.835$$

$$\rho_{max} = 0.50 \rho_b \Rightarrow \rho_{max} = 0.01063 \Rightarrow A_{smax} = \rho_{max} b d \Rightarrow A_{smax} = 58.465$$

$$\rho_d = \frac{w f_c}{f_y} \Rightarrow \rho_d = 0.00231 \Rightarrow A_{sd} = \rho_d b d \Rightarrow A_{sd} = 12.677 \text{ cm}^2$$

$$A_s \phi 5/8: d_b \phi 5/8 = 1.588 \text{ cm} \Rightarrow \phi 5/8 = 2.00 \text{ cm}^2$$

$$\# \text{ DE VARILLAS: } A_{std} = A_{sd} \times B_y = 13.284 (3.00) \Rightarrow A_{std} = 39.852 \text{ cm}^2$$

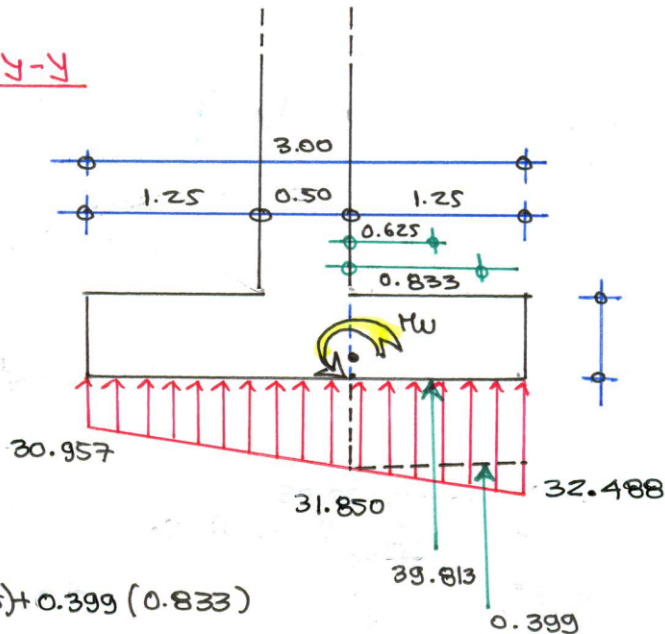
$$N = \frac{A_{std}}{A_{\phi}} \Rightarrow N = \frac{39.852}{2.00} \Rightarrow N = 20 \text{ UNIDADES}$$

DISTRIBUCIÓN DE ACERO

$$S = \frac{B_x - 2r - \phi - 0.10}{N-1} \Rightarrow S = \frac{3.00 - 0.015 - 1.588 - 0.10}{19} \Rightarrow S = 144 \text{ mm} \sim 0.15$$

19 $\phi 5/8$ @ 0.15

DIRECCION : Y-Y



$$W = 39.813(0.625) + 0.399(0.833)$$

$$W = 25.215 \text{ ton-m.}$$

DATOS: $W = 26.37 \text{ ton-m}$, $b = 100 \text{ cm}$, $d = 55 \text{ cm}$, $f_c = 210 \text{ kg/cm}^2$, $f_y = 4200 \text{ kg/cm}^2$

$$\rho_{\min} = 0.70 \sqrt{f_c} / f_y$$

$$\Rightarrow \rho_{\min} = 0.00242$$

$$A_{s\min} = 13.284 \text{ cm}^2$$

$$\rho_b = 0.85 \beta_1 \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow \rho_b = 0.02125$$

$$A_{sb} = 116.875 \text{ cm}^2$$

$$\rho_d = \omega \frac{f_c}{f_y}$$

$$\Rightarrow \rho_d = 0.00227$$

$$A_{sd} = 12.462 \text{ cm}^2$$

$$\rho_{\max} = 0.50 \rho_b$$

$$\Rightarrow \rho_{\max} = 0.01063$$

$$A_{s\max} = 58.4375 \text{ cm}^2$$

$$A_{s\phi 5/8} = 2.00 \text{ cm}^2 \quad d_b = 0.016 \text{ cm.}$$

VARILLAS: $A_{std} = A_{sd} (B_y) \Rightarrow A_{std} = (13.284)(3) \Rightarrow A_{std} = 39.852 \text{ cm}^2$

$$N = \frac{A_{std}}{A_{\phi}} = \frac{39.852}{2.00} \Rightarrow N = 20 \text{ varillas}$$

ESPACIAMIENTO DE ACEROS

$$S = \frac{B_x - 2r - \phi - 0.10}{N-1} = \frac{3.00 - 0.15 - 0.016 - 0.10}{19} \Rightarrow S = 0.145 \text{ m.}$$

$$\phi 20 \phi 5/8" @ 0.145$$

DETALLE
EN PLANTA

